

# Scoreboard

- **Structural:** If required FU is busy then stall in **IS** stage.
- **WAW:** If an active instruction has same destination register as current instruction then stall in **IS** stage until that instruction is active.
- **RAW:** If the source registers of current instruction match to any active instruction's destination register then stall in **RO** stage until that instruction is active.
- **WAR:** If the destination register listed in any of the active instruction's source registers then stall in **WR** until that instruction read its operands

# Scoreboard

## **Execution pipeline lengths:**

- FP ADD/SUB=2cc
- FP MUL=10cc
- FP DIV= 20cc
- L.D/S.D=1 cc

## **Functional Units:**

- FP MUL=1
- FP DIV=1
- FP ADD=1
- INT=1

# Scoreboard

FP ADD/SUB=2 cc, FP MUL=10 cc, FP DIV=20 cc, L.D/S.D: 1 cc  
FP MUL=1, FP DIV=1, FP ADD=1, INT=1

| inst               | IS | RO | EX | WR | Comments |
|--------------------|----|----|----|----|----------|
| L.D     F4, 0(R2)  |    |    |    |    |          |
| DIV.D   F5, F4, F2 |    |    |    |    |          |
| S.D     F2, 20(R2) |    |    |    |    |          |
| MUL.D   F2, F5, F0 |    |    |    |    |          |
| SUB.D   F0, F4, F6 |    |    |    |    |          |
| ADD.D   F0, F0, F2 |    |    |    |    |          |

# Scoreboard

FP ADD/SUB=2 cc, FP MUL=10 cc, FP DIV=20 cc, L.D/S.D: 1 cc  
FP MUL=1, FP DIV=1, FP ADD=1, INT=1

| inst               | IS | RO | EX | WR | Comments |
|--------------------|----|----|----|----|----------|
| L.D     F4, 0(R2)  | 1  | 2  | 3  | 4  |          |
| DIV.D   F5, F4, F2 |    |    |    |    |          |
| S.D     F2, 20(R2) |    |    |    |    |          |
| MUL.D   F2, F5, F0 |    |    |    |    |          |
| SUB.D   F0, F4, F6 |    |    |    |    |          |
| ADD.D   F0, F0, F2 |    |    |    |    |          |

# Scoreboard

FP ADD/SUB=2 cc, FP MUL=10 cc, FP DIV=20 cc, L.D/S.D: 1 cc  
FP MUL=1, FP DIV=1, FP ADD=1, INT=1

| inst               | IS | RO  | EX   | WR | Comments |
|--------------------|----|-----|------|----|----------|
| L.D    F4, 0(R2)   | 1  | 2   | 3    | 4  |          |
| DIV.D   F5, F4, F2 | 2  | 3-5 | 6-25 | 26 | RAW(F4)  |
| S.D    F2, 20(R2)  |    |     |      |    |          |
| MUL.D   F2, F5, F0 |    |     |      |    |          |
| SUB.D   F0, F4, F6 |    |     |      |    |          |
| ADD.D   F0, F0, F2 |    |     |      |    |          |

# Scoreboard

FP ADD/SUB=2 cc, FP MUL=10 cc, FP DIV=20 cc, L.D/S.D: 1 cc  
FP MUL=1, FP DIV=1, FP ADD=1, INT=1

| inst               | IS  | RO  | EX   | WR | Comments    |
|--------------------|-----|-----|------|----|-------------|
| L.D    F4, 0(R2)   | 1   | 2   | 3    | 4  |             |
| DIV.D   F5, F4, F2 | 2   | 3-5 | 6-25 | 26 | RAW(F4)     |
| S.D    F2, 20(R2)  | 3-5 | 6   | 7    | 8  | Struct(INT) |
| MUL.D   F2, F5, F0 |     |     |      |    |             |
| SUB.D   F0, F4, F6 |     |     |      |    |             |
| ADD.D   F0, F0, F2 |     |     |      |    |             |

# Scoreboard

FP ADD/SUB=2 cc, FP MUL=10 cc, FP DIV=20 cc, L.D/S.D: 1 cc  
FP MUL=1, FP DIV=1, FP ADD=1, INT=1

| inst               | IS  | RO   | EX    | WR | Comments    |
|--------------------|-----|------|-------|----|-------------|
| L.D    F4, 0(R2)   | 1   | 2    | 3     | 4  |             |
| DIV.D   F5, F4, F2 | 2   | 3-5  | 6-25  | 26 | RAW(F4)     |
| S.D    F2, 20(R2)  | 3-5 | 6    | 7     | 8  | Struct(INT) |
| MUL.D   F2, F5, F0 | 6   | 7-27 | 28-37 | 38 | RAW(F5)     |
| SUB.D   F0, F4, F6 |     |      |       |    |             |
| ADD.D   F0, F0, F2 |     |      |       |    |             |

# Scoreboard

FP ADD/SUB=2 cc, FP MUL=10 cc, FP DIV=20 cc, L.D/S.D: 1 cc  
FP MUL=1, FP DIV=1, FP ADD=1, INT=1

| inst               | IS  | RO   | EX    | WR    | Comments    |
|--------------------|-----|------|-------|-------|-------------|
| L.D    F4, 0(R2)   | 1   | 2    | 3     | 4     |             |
| DIV.D   F5, F4, F2 | 2   | 3-5  | 6-25  | 26    | RAW(F4)     |
| S.D    F2, 20(R2)  | 3-5 | 6    | 7     | 8     | Struct(INT) |
| MUL.D   F2, F5, F0 | 6   | 7-27 | 28-37 | 38    | RAW(F5)     |
| SUB.D   F0, F4, F6 | 7   | 8    | 9-10  | 11-28 | WAR(F0)     |
| ADD.D   F0, F0, F2 |     |      |       |       |             |

# Scoreboard

FP ADD/SUB=2 cc, FP MUL=10 cc, FP DIV=20 cc, L.D/S.D: 1 cc  
FP MUL=1, FP DIV=1, FP ADD=1, INT=1

| inst               | IS   | RO    | EX    | WR    | Comments                         |
|--------------------|------|-------|-------|-------|----------------------------------|
| L.D    F4, 0(R2)   | 1    | 2     | 3     | 4     |                                  |
| DIV.D   F5, F4, F2 | 2    | 3-5   | 6-25  | 26    | RAW(F4)                          |
| S.D    F2, 20(R2)  | 3-5  | 6     | 7     | 8     | Struct(INT)                      |
| MUL.D   F2, F5, F0 | 6    | 7-27  | 28-37 | 38    | RAW(F5)                          |
| SUB.D   F0, F4, F6 | 7    | 8     | 9-10  | 11-28 | WAR(F0)                          |
| ADD.D   F0, F0, F2 | 8-29 | 30-39 | 40-41 | 42    | Struct(ADD), WAW(F0),<br>RAW(F2) |

# Scoreboard

| inst  |            | IS   | RO    | EX    | WR    | Comments                         |
|-------|------------|------|-------|-------|-------|----------------------------------|
| L.D   | F4, 0(R2)  | 1    | 2     | 3     | 4     |                                  |
| DIV.D | F5, F4, F2 | 2    | 3-5   | 6-25  | 26    | RAW(F4)                          |
| S.D   | F2, 20(R2) | 3-5  | 6     | 7     | 8     | Struct(INT)                      |
| MUL.D | F2, F5, F0 | 6    | 7-27  | 28-37 | 38    | RAW(F5)                          |
| SUB.D | F0, F4, F6 | 7    | 8     | 9-10  | 11-28 | WAR(F0)                          |
| ADD.D | F0, F0, F2 | 8-29 | 30-39 | 40-41 | 42    | Struct(ADD), WAW(F0),<br>RAW(F2) |

### Register status table when SUB.D enters RO stage?

[illegible]

# Scoreboard

| inst  |            | IS   | RO    | EX    | WR    | Comments                         |
|-------|------------|------|-------|-------|-------|----------------------------------|
| L.D   | F4, 0(R2)  | 1    | 2     | 3     | 4     |                                  |
| DIV.D | F5, F4, F2 | 2    | 3-5   | 6-25  | 26    | RAW(F4)                          |
| S.D   | F2, 20(R2) | 3-5  | 6     | 7     | 8     | Struct(INT)                      |
| MUL.D | F2, F5, F0 | 6    | 7-27  | 28-37 | 38    | RAW(F5)                          |
| SUB.D | F0, F4, F6 | 7    | 8     | 9-10  | 11-28 | WAR(F0)                          |
| ADD.D | F0, F0, F2 | 8-29 | 30-39 | 40-41 | 42    | Struct(ADD), WAW(F0),<br>RAW(F2) |

### Register status table when SUB.D enters RO stage?

[illegible]

# Scoreboard

INT/S.D/L.D = 1 cc, MUL.D = 10 cc, DIV.D = 20 cc, ADD.D/SUB.D = 2 cc  
1 MUL.D unit, 2 ADD.D/SUB.D units, 1 DIV.D unit, 2 INT unit

| Instructions          | IS | RO | EX | WR | Comments |
|-----------------------|----|----|----|----|----------|
| L.D      F2, 0 (R3)   | 1  | 2  | 3  | 4  |          |
| ADD.D    F5, F2, F1   |    |    |    |    |          |
| MUL.D    F10, F5, F8  |    |    |    |    |          |
| DIV.D    F9, F2, F12  |    |    |    |    |          |
| SUB.D    F8, F10, F6  |    |    |    |    |          |
| S.D       F8, 0 (R3)  |    |    |    |    |          |
| L.D       F8, 45 (R2) |    |    |    |    |          |
| DADDI    R3, R3, 8    |    |    |    |    |          |

# Scoreboard

INT/S.D/L.D = 1 cc, MUL.D = 10 cc, DIV.D = 20 cc, ADD.D/SUB.D = 2 cc  
1 MUL.D unit, 2 ADD.D/SUB.D units, 1 DIV.D unit, 2 INT unit

| Instructions          | IS | RO  | EX  | WR | Comments |
|-----------------------|----|-----|-----|----|----------|
| L.D      F2, 0 (R3)   | 1  | 2   | 3   | 4  |          |
| ADD.D    F5, F2, F1   | 2  | 3-5 | 6-7 | 8  | RAW(F2)  |
| MUL.D    F10, F5, F8  |    |     |     |    |          |
| DIV.D    F9, F2, F12  |    |     |     |    |          |
| SUB.D    F8, F10, F6  |    |     |     |    |          |
| S.D       F8, 0 (R3)  |    |     |     |    |          |
| L.D       F8, 45 (R2) |    |     |     |    |          |
| DADDI    R3, R3, 8    |    |     |     |    |          |

# Scoreboard

INT/S.D/L.D = 1 cc, MUL.D = 10 cc, DIV.D = 20 cc, ADD.D/SUB.D = 2 cc  
1 MUL.D unit, 2 ADD.D/SUB.D units, 1 DIV.D unit, 2 INT unit

| Instructions         | IS | RO  | EX    | WR | Comments |
|----------------------|----|-----|-------|----|----------|
| L.D      F2, 0 (R3)  | 1  | 2   | 3     | 4  |          |
| ADD.D    F5, F2, F1  | 2  | 3-5 | 6-7   | 8  | RAW(F2)  |
| MUL.D    F10, F5, F8 | 3  | 4-9 | 10-19 | 20 | RAW(F5)  |
| DIV.D    F9, F2, F12 |    |     |       |    |          |
| SUB.D    F8, F10, F6 |    |     |       |    |          |
| S.D      F8, 0 (R3)  |    |     |       |    |          |
| L.D      F8, 45 (R2) |    |     |       |    |          |
| DADDI    R3, R3, 8   |    |     |       |    |          |

# Scoreboard

INT/S.D/L.D = 1 cc, MUL.D = 10 cc, DIV.D = 20 cc, ADD.D/SUB.D = 2 cc  
1 MUL.D unit, 2 ADD.D/SUB.D units, 1 DIV.D unit, 2 INT unit

| Instructions          | IS | RO  | EX    | WR | Comments |
|-----------------------|----|-----|-------|----|----------|
| L.D      F2, 0 (R3)   | 1  | 2   | 3     | 4  |          |
| ADD.D    F5, F2, F1   | 2  | 3-5 | 6-7   | 8  | RAW(F2)  |
| MUL.D    F10, F5, F8  | 3  | 4-9 | 10-19 | 20 | RAW(F5)  |
| DIV.D    F9, F2, F12  | 4  | 5   | 6-25  | 26 |          |
| SUB.D    F8, F10, F6  |    |     |       |    |          |
| S.D       F8, 0 (R3)  |    |     |       |    |          |
| L.D       F8, 45 (R2) |    |     |       |    |          |
| DADDI    R3, R3, 8    |    |     |       |    |          |

# Scoreboard

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1 MUL.D unit, 2 ADD.D/SUB.D units, 1 DIV.D unit, 2 INT unit

| Instructions         | IS | RO   | EX    | WR | Comments |
|----------------------|----|------|-------|----|----------|
| L.D      F2, 0 (R3)  | 1  | 2    | 3     | 4  |          |
| ADD.D    F5, F2, F1  | 2  | 3-5  | 6-7   | 8  | RAW(F2)  |
| MUL.D    F10, F5, F8 | 3  | 4-9  | 10-19 | 20 | RAW(F5)  |
| DIV.D    F9, F2, F12 | 4  | 5    | 6-25  | 26 |          |
| SUB.D    F8, F10, F6 | 5  | 6-21 | 22-23 | 24 | RAW(F10) |
| S.D      F8, 0 (R3)  |    |      |       |    |          |
| L.D      F8, 45 (R2) |    |      |       |    |          |
| DADDI    R3, R3, 8   |    |      |       |    |          |

# Scoreboard

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1 MUL.D unit, 2 ADD.D/SUB.D units, 1 DIV.D unit, 2 INT unit

| Instructions         | IS | RO   | EX    | WR | Comments |
|----------------------|----|------|-------|----|----------|
| L.D      F2, 0 (R3)  | 1  | 2    | 3     | 4  |          |
| ADD.D    F5, F2, F1  | 2  | 3-5  | 6-7   | 8  | RAW(F2)  |
| MUL.D    F10, F5, F8 | 3  | 4-9  | 10-19 | 20 | RAW(F5)  |
| DIV.D    F9, F2, F12 | 4  | 5    | 6-25  | 26 |          |
| SUB.D    F8, F10, F6 | 5  | 6-21 | 22-23 | 24 | RAW(F10) |
| S.D      F8, 0 (R3)  | 6  | 7-25 | 26    | 27 | RAW(F8)  |
| L.D      F8, 45 (R2) |    |      |       |    |          |
| DADDI    R3, R3, 8   |    |      |       |    |          |

# Scoreboard

INT/S.D/L.D = 1 cc, MUL.D = 10 cc, DIV.D = 20 cc, ADD.D/SUB.D = 2 cc  
1 MUL.D unit, 2 ADD.D/SUB.D units, 1 DIV.D unit, 2 INT unit

| Instructions          | IS   | RO   | EX    | WR | Comments |
|-----------------------|------|------|-------|----|----------|
| L.D      F2, 0 (R3)   | 1    | 2    | 3     | 4  |          |
| ADD.D    F5, F2, F1   | 2    | 3-5  | 6-7   | 8  | RAW(F2)  |
| MUL.D    F10, F5, F8  | 3    | 4-9  | 10-19 | 20 | RAW(F5)  |
| DIV.D    F9, F2, F12  | 4    | 5    | 6-25  | 26 |          |
| SUB.D    F8, F10, F6  | 5    | 6-21 | 22-23 | 24 | RAW(F10) |
| S.D       F8, 0 (R3)  | 6    | 7-25 | 26    | 27 | RAW(F8)  |
| L.D       F8, 45 (R2) | 7-25 | 26   | 27    | 28 | WAW(F8)  |
| DADDI    R3, R3, 8    |      |      |       |    |          |

# Scoreboard

INT/S.D/L.D = 1 cc, MUL.D = 10 cc, DIV.D = 20 cc, ADD.D/SUB.D = 2 cc  
1 MUL.D unit, 2 ADD.D/SUB.D units, 1 DIV.D unit, 2 INT unit

| Instructions         | IS    | RO   | EX    | WR | Comments    |
|----------------------|-------|------|-------|----|-------------|
| L.D      F2, 0 (R3)  | 1     | 2    | 3     | 4  |             |
| ADD.D    F5, F2, F1  | 2     | 3-5  | 6-7   | 8  | RAW(F2)     |
| MUL.D    F10, F5, F8 | 3     | 4-9  | 10-19 | 20 | RAW(F5)     |
| DIV.D    F9, F2, F12 | 4     | 5    | 6-25  | 26 |             |
| SUB.D    F8, F10, F6 | 5     | 6-21 | 22-23 | 24 | RAW(F10)    |
| S.D      F8, 0 (R3)  | 6     | 7-25 | 26    | 27 | RAW(F8)     |
| L.D      F8, 45 (R2) | 7-25  | 26   | 27    | 28 | WAW(F8)     |
| DADDI    R3, R3, 8   | 26-28 | 29   | 30    | 31 | Struct(INT) |

# Scoreboard

| Instructions          | IS    | RO   | EX    | WR | Comments    |
|-----------------------|-------|------|-------|----|-------------|
| L.D      F2, 0 (R3)   | 1     | 2    | 3     | 4  |             |
| ADD.D    F5, F2, F1   | 2     | 3-5  | 6-7   | 8  | RAW(F2)     |
| MUL.D    F10, F5, F8  | 3     | 4-9  | 10-19 | 20 | RAW(F5)     |
| DIV.D    F9, F2, F12  | 4     | 5    | 6-25  | 26 |             |
| SUB.D    F8, F10, F6  | 5     | 6-21 | 22-23 | 24 | RAW(F10)    |
| S.D       F8, 0 (R3)  | 6     | 7-25 | 26    | 27 | RAW(F8)     |
| L.D       F8, 45 (R2) | 7-25  | 26   | 27    | 28 | WAW(F8)     |
| DADDI    R3, R3, 8    | 26-28 | 29   | 30    | 31 | Struct(INT) |

**State of functional units at the end of 15<sup>th</sup> clock cycle**

| Functional Unit Status |       |      |     |             |              |              |                 |                 |          |             |
|------------------------|-------|------|-----|-------------|--------------|--------------|-----------------|-----------------|----------|-------------|
| No.                    | Name  | Busy | Op  | Fi<br>(Dst) | Fj<br>(Src1) | Fk<br>(Src2) | Qj<br>(Src1 FU) | Qk<br>(Src2 FU) | Rj (Flg) | Rk<br>(Flg) |
| 1                      | INT   |      |     |             |              |              |                 |                 |          |             |
| 2                      | INT   | YES  | S.D |             | F8           | R3           | FPADD(4)        |                 | 0        | 1           |
| 3                      | FPADD |      |     |             |              |              |                 |                 |          |             |
| 4                      | FPADD | YES  | SUB | F8          | F10          | F6           | FPMUL(6)        |                 | 0        | 1           |
| 5                      | FPDIV | YES  | DIV | F9          | F2           | F12          |                 |                 | 0        | 0           |
| 6                      | FPMUL | YES  | MUL | F10         | F5           | F8           |                 |                 | 0        | 0           |